

Libraries required

The following libraries must either be installed globally in your python directory (i.e. anaconda, miniconda) or in the python environment being used to use srimpro. This should be done automatically when installing srimpro, but can be done manually if for some reason the dependencies are not properly installed. These libraries can be installed using the listed commands, either from Conda, PyPi, or docker.

- Numpy
 - `conda install numpy`
 - `pip install numpy`
- Pandas
 - `conda install pandas`
 - `pip install pandas`
- Openpyxl
 - `conda install anaconda::openpyxl`
- Pysrim
 - `conda install -c costrouc pysrim`
 - `pip install pysrim`
 - `docker pull costrouc/pysrim`

Modified pysrim files to replace

All srimpro functions use the pysrim library to read SRIM output files before performing calculations. However, the latest version of this library is not compatible with the python versions 3.12 and beyond. The srimpro repository contains updated versions of certain pysrim files that make it compatible with python 3.14 (the latest version of python as of the release of srimpro), with additional minor modifications to allow for increased computational efficiency in srimpro. These incompatibilities with current versions of python were pointed out by comments posted in the Issues tab of the pysrim GitHub. Specifically, the the yaml loader issue was fixed, and the use of distutils was replaced by shutils. These files are available in the srimpro repository in `srimpro/pysrim_updates`. After installing pysrim, the following instructions must be followed to replace the outdated pysrim files.

1. Download the three files within `srimpro/pysrim_updates`, these will be used to replace their corresponding pysrim files
2. Using Windows Powershell, Anaconda Prompt, or equivalent, run the command `pip show srims` to find the exact directory in which pysrim is installed
3. Copy/cut the pysrim_updates files, then navigate to this directory in file explorer and paste the copied files into the srims folder
4. Delete the original versions of these files from pysrim
 - a. In the main srims folder, delete `output.py`
 - b. In the main srims folder, delete `srims.py`

- c. In the core subfolder, delete elementdb.py
- 5. Rename the pysrim_updated files to the names of their corresponding pysrim files that were just deleted (i.e. remove the “_updated” from the end of the name of each file)
- 6. Move the new elementdb.py file to the core folder